



School of Computer Science Engineering and Information Systems
Fall Semester 2026-2027

Assessment 1

**Course Name & code: Agile Development Process
and Devops -ISWE406L**

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Lab-1

PYTHON CODE

A screenshot of a code editor window titled "calculator.py X". The editor shows the following Python code:

```
1 # calculator.py
2 # A simple calculator program
3 def add(x, y):
4     return x + y
5 def subtract(a, b):
6     return a - b
7 if __name__ == "__main__":
8     x = 10
9     y = 5
10    print("Sum:", add(x, y))
11    print("Difference:", subtract(x, y))
```

The left sidebar shows a file explorer with "calculator.py" and "grades.py" under a "GITHUB" folder. The bottom panel shows a terminal with the command "python calculator.py" and its output: "Sum: 15" and "Difference: 5".

Initialize the repository

Commit the file

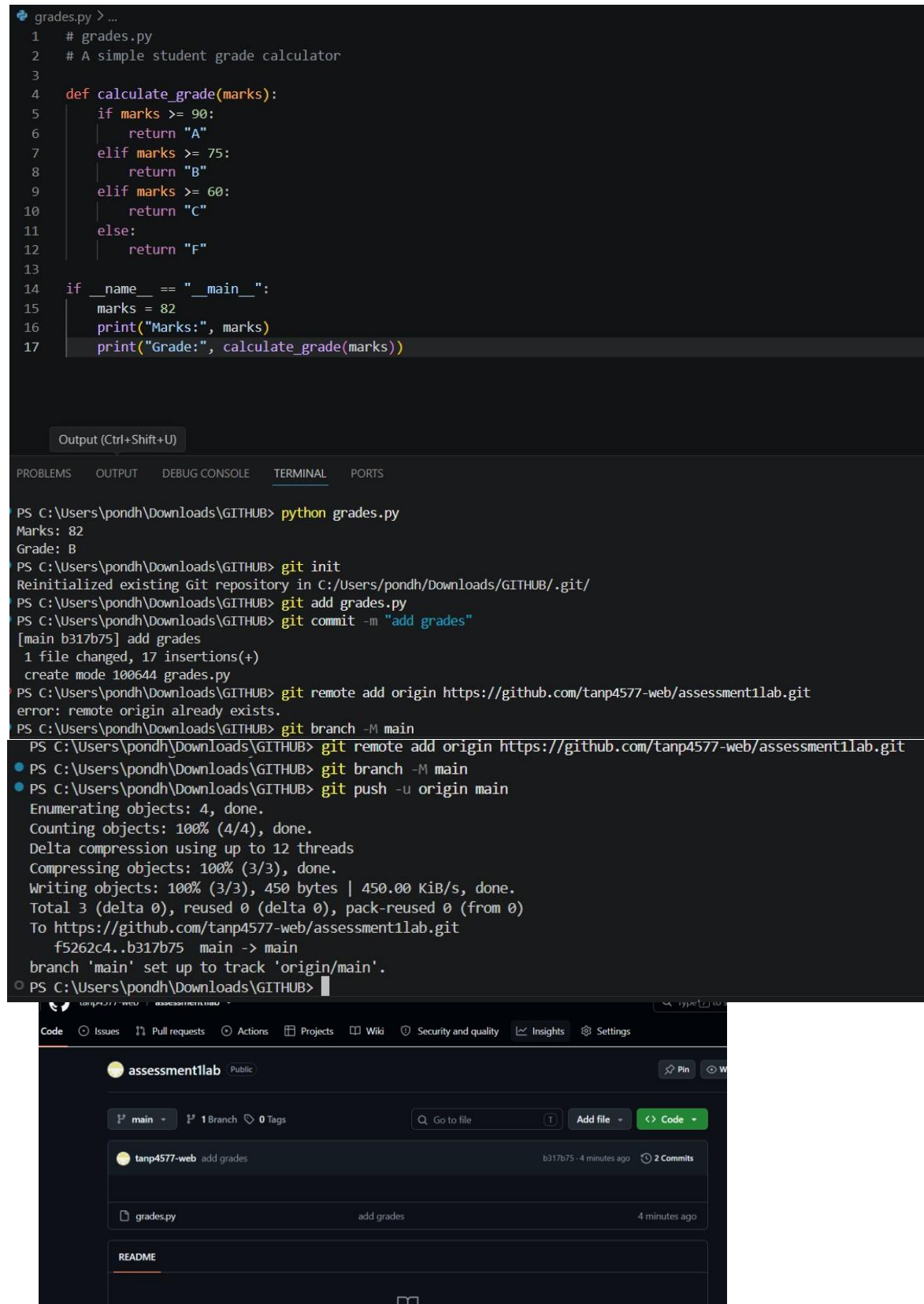
Link the local repository to GitHub

```
PS C:\Users\pondh\Downloads\GITHUB> python calculator.py
Sum: 15
Difference: 5
PS C:\Users\pondh\Downloads\GITHUB> git init
Initialized empty Git repository in C:/Users/pondh/Downloads/GITHUB/.git/
PS C:\Users\pondh\Downloads\GITHUB> git add calculator.py
PS C:\Users\pondh\Downloads\GITHUB> git commit -m "add calculator"
[master (root-commit) f5262c4] add calculator
 1 file changed, 11 insertions(+)
 create mode 100644 calculator.py
PS C:\Users\pondh\Downloads\GITHUB> git remote add origin https://github.com/tanp4577-web/assessment1lab.git
```

Rename the branch and push

```
PS C:\Users\pondh\Downloads\GITHUB> git branch -M main
PS C:\Users\pondh\Downloads\GITHUB> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 370 bytes | 370.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/tanp4577-web/assessment1lab.git
```

LAB-2 BRANCHING



The image shows a VS Code editor window with a Python file named `grades.py` and a terminal window below it.

grades.py

```
1 # grades.py
2 # A simple student grade calculator
3
4 def calculate_grade(marks):
5     if marks >= 90:
6         return "A"
7     elif marks >= 75:
8         return "B"
9     elif marks >= 60:
10        return "C"
11    else:
12        return "F"
13
14 if __name__ == "__main__":
15     marks = 82
16     print("Marks:", marks)
17     print("Grade:", calculate_grade(marks))
```

Terminal Output

```
PS C:\Users\pondh\Downloads\GITHUB> python grades.py
Marks: 82
Grade: B
PS C:\Users\pondh\Downloads\GITHUB> git init
Reinitialized existing Git repository in C:/Users/pondh/Downloads/GITHUB/.git/
PS C:\Users\pondh\Downloads\GITHUB> git add grades.py
PS C:\Users\pondh\Downloads\GITHUB> git commit -m "add grades"
[main b317b75] add grades
1 file changed, 17 insertions(+)
create mode 100644 grades.py
PS C:\Users\pondh\Downloads\GITHUB> git remote add origin https://github.com/tanp4577-web/assessment1lab.git
error: remote origin already exists.
PS C:\Users\pondh\Downloads\GITHUB> git branch -M main
PS C:\Users\pondh\Downloads\GITHUB> git remote add origin https://github.com/tanp4577-web/assessment1lab.git
PS C:\Users\pondh\Downloads\GITHUB> git branch -M main
PS C:\Users\pondh\Downloads\GITHUB> git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 450 bytes | 450.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/tanp4577-web/assessment1lab.git
f5262c4..b317b75 main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\pondh\Downloads\GITHUB>
```

GitHub Repository View

The screenshot shows the GitHub repository page for `assessment1lab`. The repository is public and has 1 branch (main) and 0 tags. The commit history shows a single commit by `tanp4577-web` titled "add grades" at commit `b317b75`, made 4 minutes ago. The file `grades.py` is listed with the same commit. The README section is visible at the bottom.

1. git branch — list branches

```
PS C:\Users\pondh\Downloads\GITHUB> git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 450 bytes | 450.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/tanp4577-web/assessment1lab.git
   f5262c4..b317b75  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\pondh\Downloads\GITHUB> git branch
* main
```

2. git branch — create a branch

```
branch 'main' set up to track 'origin/main'.
PS C:\Users\pondh\Downloads\GITHUB> git branch
* main
PS C:\Users\pondh\Downloads\GITHUB> git branch feature-average
PS C:\Users\pondh\Downloads\GITHUB> git branch
feature-average
* main
PS C:\Users\pondh\Downloads\GITHUB>
```

3. git switch — move to a branch

```
PS C:\Users\pondh\Downloads\GITHUB> git switch branch feature-average
fatal: only one reference expected
PS C:\Users\pondh\Downloads\GITHUB> git switch feature-average
Switched to branch 'feature-average'
PS C:\Users\pondh\Downloads\GITHUB>
```

```
grades.py > ...
1  # grades.py
2  # A simple student grade calculator
3
4  def calculate_grade(marks):
5      if marks >= 90:
6          return "A"
7      elif marks >= 75:
8          return "B"
9      elif marks >= 60:
10         return "C"
11     else:
12         return "F"
13     def average(marks_list):
14         return sum(marks_list) / len(marks_list)
15
16 if __name__ == "__main__":
17     marks = 82
18     print("Marks:", marks)
19     print("Grade:", calculate_grade(marks))

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\pondh\Downloads\GITHUB> python -u "c:\Users\pondh\Downloads\GITHUB\grades.py"
Marks: 82
Grade: B
PS C:\Users\pondh\Downloads\GITHUB> python grades.py
Marks: 82
Grade: B
PS C:\Users\pondh\Downloads\GITHUB> git add grades.py
PS C:\Users\pondh\Downloads\GITHUB> git commit -m "Add average Function"
[feature-average cc2e4b7] Add average Function
1 file changed, 2 insertions(+)
PS C:\Users\pondh\Downloads\GITHUB> 
```

4. git switch main — see that branches are isolated

```
• PS C:\Users\pondh\Downloads\GITHUB> git add grades.py
• PS C:\Users\pondh\Downloads\GITHUB> git commit -m "Add average Function"
[feature-average cc2e4b7] Add average Function
1 file changed, 2 insertions(+)
• PS C:\Users\pondh\Downloads\GITHUB> git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
• PS C:\Users\pondh\Downloads\GITHUB> 
```


5. `git switch -c` — create and switch in one command

```
def topper(marks_dict):  
    return max(marks_dict, key=marks_dict.get)
```

```
PS C:\Users\pondh\Downloads\GITHUB> git switch main  
Switched to branch 'main'  
Your branch is up to date with 'origin/main'.  
PS C:\Users\pondh\Downloads\GITHUB> git switch -c feature-topper  
Switched to a new branch 'feature-topper'  
PS C:\Users\pondh\Downloads\GITHUB> git add grades.py  
PS C:\Users\pondh\Downloads\GITHUB> git commit -m "ADD TOPPER FEATURE"  
On branch feature-topper  
nothing to commit, working tree clean  
PS C:\Users\pondh\Downloads\GITHUB>
```

6. `git branch -v`

```
PS C:\Users\pondh\Downloads\GITHUB> git branch -v  
feature-average cc2e4b7 Add average Function  
* feature-topper b317b75 add grades  
main b317b75 add grades  
PS C:\Users\pondh\Downloads\GITHUB>
```

7. `git merge` — bring a branch into main

```
PS C:\Users\pondh\Downloads\GITHUB> git switch main  
Switched to branch 'main'  
Your branch is up to date with 'origin/main'.  
PS C:\Users\pondh\Downloads\GITHUB> git merge feature-average  
Updating b317b75..cc2e4b7  
Fast-forward  
grades.py | 2 ++  
1 file changed, 2 insertions(+)  
PS C:\Users\pondh\Downloads\GITHUB> git merge feature-topper  
Already up to date.  
PS C:\Users\pondh\Downloads\GITHUB>
```

8. git log --graph — visualize the branch history

```
PS C:\Users\pondh\Downloads\GITHUB> git log --oneline --graph --all
>>
* cc2e4b7 (HEAD -> main, feature-average) Add average Function
* b317b75 (origin/main, feature-topper) add grades
* f5262c4 add calculator
PS C:\Users\pondh\Downloads\GITHUB> █
```

9. git push — send branches to GitHub

```
PS C:\Users\pondh\Downloads\GITHUB> git push
PS C:\Users\pondh\Downloads\GITHUB> git switch feature-average
Switched to branch 'feature-average'
PS C:\Users\pondh\Downloads\GITHUB> git push -u origin feature-average
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-average' on GitHub by visiting:
remote:   https://github.com/tanp4577-web/assessment1lab/pull/new/feature-average
remote:
To https://github.com/tanp4577-web/assessment1lab.git
 * [new branch]      feature-average -> feature-average
branch 'feature-average' set up to track 'origin/feature-average'.
PS C:\Users\pondh\Downloads\GITHUB> █
```

10. git branch -m — rename a branch

```
PS C:\Users\pondh\Downloads\GITHUB> git branch -v
* feature-average cc2e4b7 Add average Function
  feature-avg     cc2e4b7 Add average Function
  feature-topper  b317b75 add grades
  main           cc2e4b7 Add average Function
PS C:\Users\pondh\Downloads\GITHUB> █
```

11. git branch -d — delete a merged branch

```
PS C:\Users\pondh\Downloads\GITHUB> git branch -d feature-avg
Deleted branch feature-avg (was cc2e4b7).
PS C:\Users\pondh\Downloads\GITHUB> git branch -d feature-topper
Deleted branch feature-topper (was b317b75).
PS C:\Users\pondh\Downloads\GITHUB> █
```

12. git branch -D — force delete an unmerged branch

```
#TANMAY SATISH PONDHE 24MIS0318|
```

```
PS C:\Users\pondh\Downloads\GITHUB> git switch -c experiment
PS C:\Users\pondh\Downloads\GITHUB> git add grades.py
PS C:\Users\pondh\Downloads\GITHUB> git commit -m "Experimental change"
[experiment b477a25] Experimental change
 1 file changed, 6 insertions(+), 4 deletions(-)
PS C:\Users\pondh\Downloads\GITHUB> git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\pondh\Downloads\GITHUB> git branch -d experiment
error: the branch 'experiment' is not fully merged
hint: If you are sure you want to delete it, run 'git branch -D experiment'
hint: Disable this message with "git config set advice.forceDeleteBranch false"
PS C:\Users\pondh\Downloads\GITHUB>
```

```
PS C:\Users\pondh\Downloads\GITHUB> git branch -D experiment
>>
Deleted branch experiment (was b477a25).
PS C:\Users\pondh\Downloads\GITHUB>
```


Lab 3-clone

```
contacts.py > ...
1 # contacts.py      Pin selection to current chat prompt (Ctrl+Alt+X) | Don't show this again (Alt+/
2 # A simple contact book program
3
4 contacts = {}
5
6
7 def add_contact(name, phone):
8     contacts[name] = phone
9
10
11 def search_contact(name):
12     return contacts.get(name, "Not found")
13
14
15 if __name__ == "__main__":
16     add_contact("Arun", "9876543210")
17     add_contact("Priya", "9123456780")
18     print("Arun's number:", search_contact("Arun"))
19     print("Search Kabir:", search_contact("Kabir"))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\pondh\Downloads\gitclonelab> cd ~\Desktop
PS C:\Users\pondh\Desktop> git clone https://github.com/tanp4577-web/gitclonelab.git
Cloning into 'gitclonelab'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
PS C:\Users\pondh\Desktop> git clone https://github.com/tanp4577-web/gitclonelab.git my-copy
Cloning into 'my-copy'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
```

2. git remote — manage the GitHub link

```
PS C:\Users\pondh\Desktop> git remote -v
fatal: not a git repository (or any of the parent directories): .git
PS C:\Users\pondh\Desktop>
>> cd gitclonelab
PS C:\Users\pondh\Desktop\gitclonelab> git remote -v
origin https://github.com/tanp4577-web/gitclonelab.git (fetch)
origin https://github.com/tanp4577-web/gitclonelab.git (push)
PS C:\Users\pondh\Desktop\gitclonelab>
```

3. .gitignore

```
PS C:\Users\pondh\Desktop\gitclonelab> git remote -v
origin https://github.com/tanp4577-web/gitclonelab.git (push)
PS C:\Users\pondh\Desktop\gitclonelab> git remote rename origin upstream
PS C:\Users\pondh\Desktop\gitclonelab> git remote -v
upstream https://github.com/tanp4577-web/gitclonelab.git (fetch)
upstream https://github.com/tanp4577-web/gitclonelab.git (push)
PS C:\Users\pondh\Desktop\gitclonelab> git remote rename upstram origin
error: No such remote: 'upstram'
PS C:\Users\pondh\Desktop\gitclonelab> git remote rename upstream origin
PS C:\Users\pondh\Desktop\gitclonelab> git remote -v
origin https://github.com/tanp4577-web/gitclonelab.git (fetch)
origin https://github.com/tanp4577-web/gitclonelab.git (push)
PS C:\Users\pondh\Desktop\gitclonelab>
```

4 .git rm — remove a tracked file

```
• PS C:\Users\pondh\Desktop\gitclonelab> cd C:\Users\pondh\Desktop\gitclonelab
• PS C:\Users\pondh\Desktop\gitclonelab> Remove-Item -Recurse -Force draft.txt -ErrorAction SilentlyContinue
• PS C:\Users\pondh\Desktop\gitclonelab> New-Item draft.txt -ItemType File -Value "temp" -Force

Directory: C:\Users\pondh\Desktop\gitclonelab

Mode                LastWriteTime         Length Name
----                -
-a---             8/1/2026 12:41 AM             4 draft.txt

• PS C:\Users\pondh\Desktop\gitclonelab> git add draft.txt
• PS C:\Users\pondh\Desktop\gitclonelab> git commit -m "Add draft file"
[main a69ddf0] Add draft file
1 file changed, 1 insertion(+)
create mode 100644 draft.txt
• PS C:\Users\pondh\Desktop\gitclonelab> git rm draft.txt
rm 'draft.txt'
• PS C:\Users\pondh\Desktop\gitclonelab> git commit -m "Remove draft file"
[main 87c8ede] Remove draft file
1 file changed, 1 deletion(-)
delete mode 100644 draft.txt
• PS C:\Users\pondh\Desktop\gitclonelab>
```

5. git mv

```
• PS C:\Users\pondh\Desktop\gitclonelab> git mv contacts.py contact_book.py
• PS C:\Users\pondh\Desktop\gitclonelab> git commit -m "Rename contacts.py to contact_book.py"
[main e6e3424] Rename contacts.py to contact_book.py
1 file changed, 0 insertions(+), 0 deletions(-)
rename contacts.py => contact_book.py (100%)
• PS C:\Users\pondh\Desktop\gitclonelab> git push
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 12 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (9/9), 913 bytes | 91.00 KiB/s, done.
Total 9 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/tanp4577-web/gitclonelab.git
1577a10..e6e3424 main -> main
• PS C:\Users\pondh\Desktop\gitclonelab>
```

6. git show

8. git reset

```

PS C:\Users\pondh\Downloads\gitclonelab> git log --oneline
1577a10 (HEAD -> main, origin/main) add contact book
PS C:\Users\pondh\Downloads\gitclonelab> git restore --staged contact.py
error: pathspec 'contact.py' did not match any file(s) known to git
PS C:\Users\pondh\Downloads\gitclonelab> git restore --staged contacts.py
PS C:\Users\pondh\Downloads\gitclonelab> git restore --staged contacts.py
PS C:\Users\pondh\Downloads\gitclonelab> git add contacts.py
PS C:\Users\pondh\Downloads\gitclonelab> git commit -m "add delete_contact function"
On branch main
Your branch is up to date with 'origin/main'.

```

9. git revert

10. git tag

11. git stash

```
PS C:\Users\pondh\Downloads\gitclonelab> cd C:\Users\pondh\Desktop\gitclonelab
PS C:\Users\pondh\Desktop\gitclonelab> cd C:\Users\pondh\Downloads\gitclonelab
PS C:\Users\pondh\Downloads\gitclonelab> git add contacts.py
PS C:\Users\pondh\Downloads\gitclonelab> git commit -m "Add delete_contact function"
On branch main
Your branch is up to date with 'origin/main'.
```

[illegible]

Welcome contacts.py D .gitignore U notes.log

contacts.py > ...
1 # contacts.py
2 # A simple contact book program
3
4 contacts = {}
5
6
7 def add_contact(name, phone):

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

- PS C:\Users\pondh\Downloads\gitclonelab> git tag v1.0
- PS C:\Users\pondh\Downloads\gitclonelab> git push origin v1.0
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 440 bytes | 440.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/tanp4577-web/gitclonelab.git
* [new tag] v1.0 -> v1.0
- PS C:\Users\pondh\Downloads\gitclonelab> git tag -a v1.1 -m "Contact book with delete feature"
- PS C:\Users\pondh\Downloads\gitclonelab> git push origin v1.1
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 182 bytes | 60.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/tanp4577-web/gitclonelab.git
* [new tag] v1.1 -> v1.1
- PS C:\Users\pondh\Downloads\gitclonelab> git stash
Saved working directory and index state WIP on main: 1577a10 add contact book
- PS C:\Users\pondh\Downloads\gitclonelab> git stash list
stash@{0}: WIP on main: 1577a10 add contact book
- PS C:\Users\pondh\Downloads\gitclonelab> git stash pop
On branch main
Your branch is up to date with 'origin/main'.

